

21	(a)	Sketch	B1	for appropriate sketch showing $y=f(x)$ reflected in the x -axis	
	(b)	Circle radius 3 centre $(-2, 0)$ and $(-5, 0)$ and $(1, 0)$ labelled	M1	for 1 of centre $(-2, 0)$ implied by drawing or label or a circle of radius 3 or intersection on the x -axis at -5 or 1 implied by drawing or labels	
			M1	for 2 of centre $(-2, 0)$ implied by drawing or label or a circle of radius 3 or intersection on the x -axis at -5 or 1 implied by drawing or labels	
	(c)	$y = \frac{4}{3}x - 25$	A1	for a fully correct answer	
			M1	for method to find gradient of OP , eg $-9 \div 12 (= -\frac{3}{4}$ or $-0.75)$ oe	
			M1	(dep) for method to find gradient of tangent, m , eg $-\frac{9}{12} \times m = -1$ or $m = \frac{4}{3}$ or $1.33\dots$	
			A1	for $y - (-9) = \frac{4}{3}(x - 12)$ or $y + 9 = \frac{4}{3}(x - 12)$ oe	$12x - 9y = 225$ $4x - 3y = 75$